

CSE460: VLSI Design

Lecture 7

Pass Transistor DC Characteristics

Recall from our previous lectures that

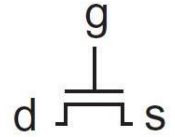
- **nMOS** transistors pass '0's well but '1's poorly
- **pMOS** transistors pass '1's well but '0's poorly

We are now better prepared to define how “poorly”

nMOS Pass Transistor DC Characteristics

To keep an nMOS transistor ON, we need

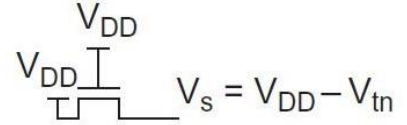
- $V_{gs} \geq V_{tn}$
- $\Rightarrow V_g - V_s \geq V_{tn}$
- $\Rightarrow V_s \leq V_g - V_{tn}$
- $\Rightarrow V_{s(max)} = V_g - V_{tn}$



nMOS Pass Transistor DC Characteristics

To keep an nMOS transistor ON, we need

- $V_{s(\max)} = V_g - V_{tn}$



nMOS transistors passing '1' ($V_{in} = V_{DD}$; $V_{out} = ?$)

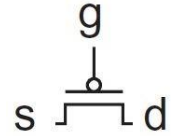
- Gate is connected to V_{DD} : $V_g = V_{DD}$
- Current flowing from drain to source
- $V_{in} = V_d = V_{DD}$; $V_{out} = V_s = ?$
- But we know $V_{s(\max)} = V_g - V_{tn} \Rightarrow V_{out(\max)} = V_{DD} - V_{tn}$

nMOS transistors attempting to pass a '1' never pass above $V_{DD} - V_{tn}$

pMOS Pass Transistor DC Characteristics

To keep a pMOS transistor ON, we need

- $V_{sg} \geq |V_{tp}|$
- $\Rightarrow V_s - V_g \geq |V_{tp}|$
- $\Rightarrow V_s \geq V_g + |V_{tp}|$
- $\Rightarrow V_{s(\min)} = V_g + |V_{tp}|$



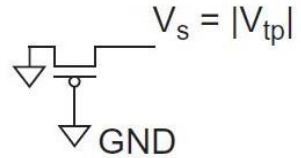
pMOS Pass Transistor DC Characteristics

To keep a pMOS transistor ON, we need

- $V_{s(\min)} = V_g + |V_{tp}|$

pMOS transistors passing '0' ($V_{in} = 0$; $V_{out} = ?$)

- Gate is connected to GND: $V_g = 0$
- Current flowing from source to drain
- $V_{in} = V_d = 0$; $V_{out} = V_s = ?$
- But we know $V_{s(\min)} = V_g + |V_{tp}| \Rightarrow V_{out(\min)} = |V_{tp}|$

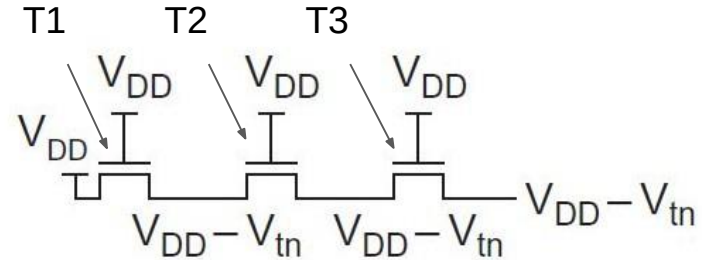


pMOS transistors attempting to pass a '0' never pass lower than $|V_{tp}|$

nMOS Pass Transistor DC Characteristics

Series nMOS transistors passing '1' (V_{DD})

- If all of the transistors have gates tied to V_{DD}
- $V_{g(T1)} = V_{g(T2)} = V_{g(T3)} = V_{DD}$
- $V_{in(T1)} = V_{DD}$; $V_{out(T3)} = ?$



1. Output of Transistor T1: $V_{out(max), T1} = V_{g(T1)} - V_{tn(T1)} = V_{DD} - V_{tn}$
2. Output of Transistor T2: $V_{out(max), T2} = V_{g(T2)} - V_{tn(T2)} = V_{DD} - V_{tn}$
3. Output of Transistor T3: $V_{out(max), T3} = V_{g(T3)} - V_{tn(T3)} = V_{DD} - V_{tn}$

nMOS Pass Transistor DC Characteristics

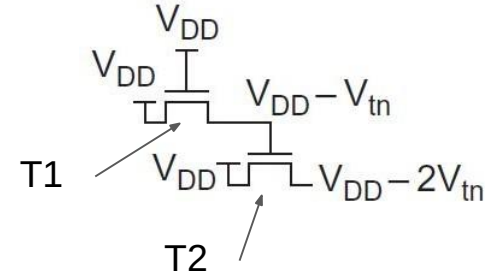
nMOS transistor driven by degraded output

- If T1 transistor has gate tied to V_{DD}
- $V_{g(T1)} = V_{DD}$; $V_{in} = V_{DD}$; $V_{out} = ?$

Output of Transistor T1: $V_{out(max), T1} = V_{g(T1)} - V_{tn(T1)} = V_{DD} - V_{tn}$

- Output of transistor T1 is now driving the gate of transistor T2
- T2 transistor has gate tied to $V_{DD} - V_{tn}$
- $V_{g(T2)} = V_{DD} - V_{tn}$; $V_{in} = V_{DD}$; $V_{out} = ?$

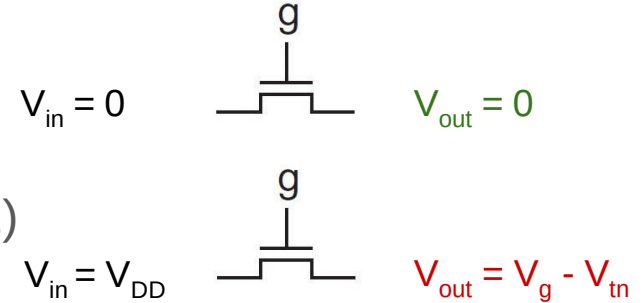
Output of Transistor T2: $V_{out(max), T2} = V_{g(T2)} - V_{tn(T2)} = (V_{DD} - V_{tn}) - V_{tn} = V_{DD} - 2V_{tn}$



Pass Transistor DC Characteristics: Summary

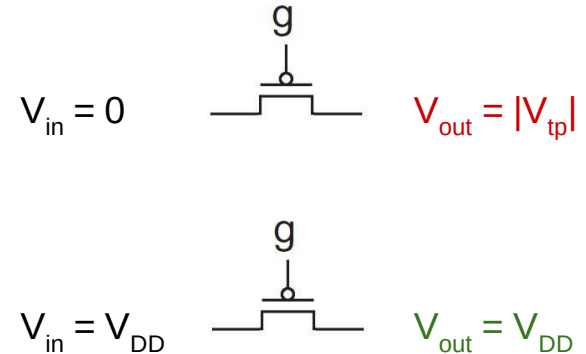
nMOS pass transistor

- $V_{in} = \text{GND} \Rightarrow V_{out} = \text{GND}$ (strong 0)
- $V_{in} = V_{DD} \Rightarrow V_{out(\max)} = V_g - V_{tn}$ (degraded 1)

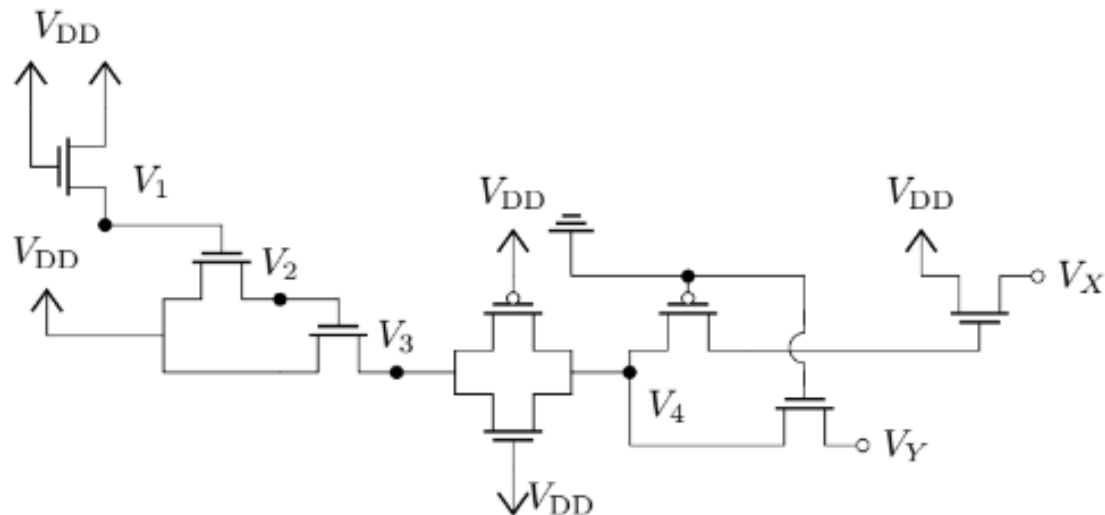


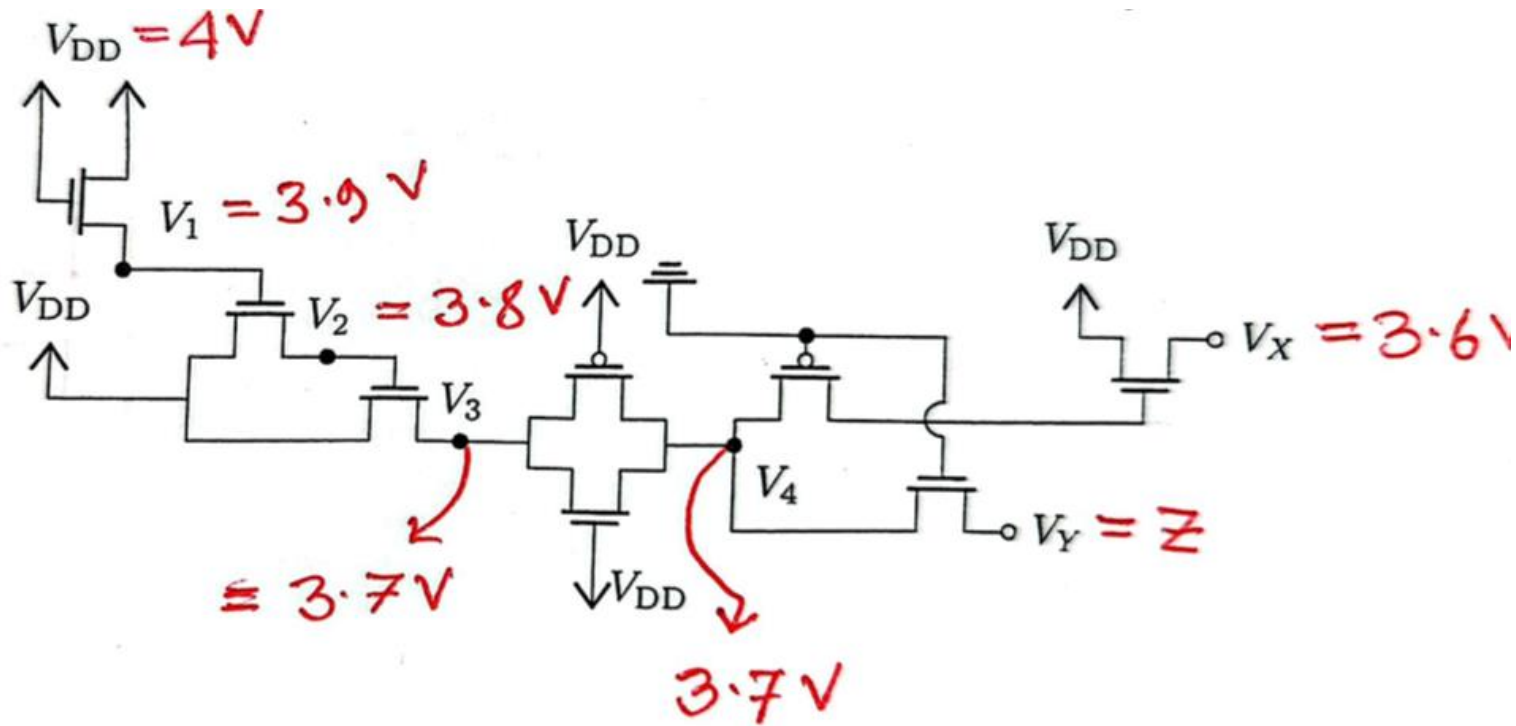
pMOS pass transistor

- $V_{in} = \text{GND} \Rightarrow V_{out(\min)} = |V_{tp}|$ (degraded 0)
- $V_{in} = V_{DD} \Rightarrow V_{out} = V_{DD}$ (strong 1)



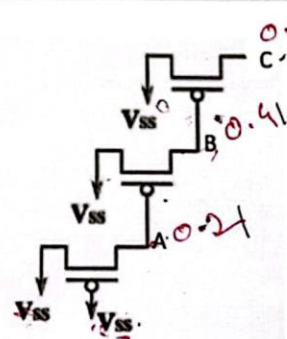
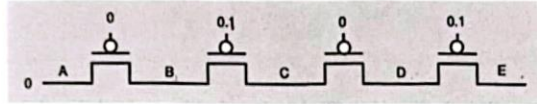
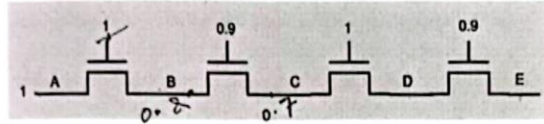
[5 marks] In the adjacent figure, the supply voltage (V_{DD}) is 4 V. The threshold voltages for the pMOS and nMOS transistors are both set at $V_{tn} = |V_{tp}| = 0.1$ V. **Determine** the voltages V_1 , V_2 , and V_4 as well as the output voltages V_X and V_Y .





Question 2: [10 Marks]

1. From the following figures, find the voltages at each of the nodes, A, B, C, D, E. Assume $V_{dd} = 3\text{ V}$, $V_{ss} = 0\text{ V}$, $V_{tn} = 0.2\text{ V}$ and $|V_{tp}| = 0.2\text{ V}$.

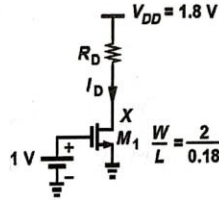


$0.6\text{ nMOS} \Rightarrow V_{GS} > V_{tn}$
 $\text{or, } V_{GS} - V_{out} > V_{tn}$
 $\text{or, } V_{out} < V_{GS} - V_{tn}$
 $(V_{out})_{max} = V_{GS} - V_{tn}$
 $\text{PMOS} \Rightarrow V_{SG} > |V_{tp}|$
 $\text{or, } V_{out} - V_{GS} > |V_{tp}|$
 $\text{or, } V_{out} > V_{GS} + |V_{tp}|$

	V_A	V_B	V_C	V_D	V_E
Circuit 1	1V 1V	0.8V	0.7V	0.7V	0.7V
Circuit 2	0V	0.2V	0.3V	0.3V	0.3V
Circuit 3	0.2V	0.4V	0.6V		

Question 1: [10 Marks]

Find the operating region of the nMOS and calculate the drain current I_D through the transistor M1 in the following figure, assuming $\mu_n C_{ox} = 100 \mu A/V^2$, $V_t = 0.4 V$. Assume that $V_D = 0.9V$. Find R_D .



$$V_{gs} = 1V, V_D = 0.9V, V_S = 0V$$

$$V_{gs} - V_{tn} = 1 - 0 - 0.4 = 0.6V$$

$$V_{DS} = 0.9 - 0 = 0.9V$$

$$V_{DS} > V_{gs} - V_{tn} \quad (\text{saturation})$$

$$\begin{aligned} \therefore I_D &= \frac{\mu_n C_{ox} \frac{W}{L}}{2} (V_{gs} - V_{tn})^2 = \frac{100 \times \frac{2}{0.18}}{2} \times 0.6^2 \\ &= 200 \mu A = 0.2mA \end{aligned}$$

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{1.8 - 0.9}{0.2} = 4.5k\Omega$$

Pass Transistor DC Characteristics: Summary

- nMOS/pMOS pass transistors sometimes produce degraded outputs
- The loss is sometimes called a threshold drop
- In old processes where V_{DD} was high and V_t was a small fraction of V_{DD} , the threshold drop was tolerable
- In modern processes V_{DD} is significantly lower and V_t is almost $\frac{1}{3}$ of V_{DD} , the threshold drop can produce an invalid or marginal logic at the output

To solve this problem we use CMOS transmission gates

Thank You!